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Studierichting: Informatica

Oefeningenreeks 1E nr. 10

$$A(z) = (z - i)(z + 2i)Q(z) + az + b$$

$$\begin{cases} A(i) = ai + b = -2 + 5i \\ A(-2i) = -2ai + b = 1 + 20i \end{cases}$$

$$\Rightarrow \begin{cases} ai + (1 + 20i + 2ai) = -2 + 5i \\ b = 1 + 20i + 2ai \end{cases}$$

$$\Rightarrow \begin{cases} 3ai = -3 - 15i \\ b = 1 + 20i + 2ai \end{cases}$$

$$\Rightarrow \begin{cases} ai = -1 - 5i \\ b = 1 + 20i + 2ai \end{cases}$$

$$\Rightarrow \begin{cases} a = -5 + i \\ b = -1 + 10i \end{cases}$$

$$\Rightarrow R(z) = (-5 + i)z - 1 + 10i$$

References